



Differences in Virginia Death Rates by Color, Sex, Age and Rural or Urban Residence
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employed in the better occupations; and a higher ratio of contact exists in residential areas where Negroes are the least concentrated. If the tenor of our interviews is to be taken seriously, however, childhood contacts with Whites probably affect the course of Negro adult life very little. Some Negroes claim that while there is much contact between White and Negro up through high school, after high school Negroes lose their White friends. This would explain why childhood contacts with Whites do not lead to more White contacts among the long resident. The effect of childhood contacts largely disappears at adulthood.

The contact of Negro adults with Whites is not, however, as entirely fortuitous as the conclusions above may appear to imply. In Table 9 we see that, for males, those who have some contact with Whites are more likely to have better jobs, better education, and higher cultural activity scores, and are more likely to vote. For females, those who have some contact with Whites are more likely to be housewives, to have higher cultural activity scores and to own their homes.

In summary, then, although the number of Negroes in the city of Madison is very small and their ratio to Whites has scarcely changed in 50 years, the Negro population remains highly transient because of meager economic opportunities. Relative freedom from "Jim Crowism" has not resulted in observable assimilation or progress. Length of residence in Madison shows little correlation with welfare.

DIFFERENCES IN VIRGINIA DEATH RATES BY COLOR, SEX, AGE AND RURAL OR URBAN RESIDENCE*

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RECENT developments in the study of vital statistics have made the factors involved in rates of death a little more apparent. Certain hypotheses have been given additional support by the new evidence available, while other generally accepted ideas are coming under the shadow of doubt. The purpose of this paper is to present some findings on this "matter of life and death" for Virginia.

A death rate is a ratio of a number of deaths, occurring in a specified period, to a given population subject to the risk of such death. In practice, there are difficulties in obtaining accurate data and in defining their use. Incomplete registration of death, misstatement of age, underenumeration of population, and limitations in classification are a few of the obstinate problems encountered in the data. In definition there is difficulty in deciding how to allocate deaths to the population subject to the risk of them; whether to relate the deaths for a stated period to the population at the beginning of the period (before the event of death) or to the number in the middle of the period; whether to make corrections in the raw data or take them as they are reported; etc. Giving a brief description of methods actually employed in this study may be the best way to clarify the definitions used and to indicate some of the limitations of the data. For the most part well established conventions have been followed.

* Manuscript received June 27, 1947.

DESCRIPTION OF METHODS

Special tabulations by place of residence of the decedent were obtained from the Virginia Bureau of Vital Statistics for the years 1939 and 1940. These tabulations were classified by color, sex, rural or urban residence and grouped by age in five-year intervals. No correction for underregistration was used.

The figures on deaths for 1939 and 1940

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were combined to get a two year average. This is generally done to overcome partially the handicap of low frequencies in some groups. The "deaths per year" used as the numerator in calculating the rate for a group was therefore an average of two years mortality experience. (Table 1 shows the

as percentages of the total deaths in the corresponding group of the county.

(2) These percentages were assumed to apply to 1939 and equivalent numbers of deaths were allocated to each urban place in that year.

The population used as the denominator

TABLE 1. VIRGINIA DEATHS BY AGE, SEX, COLOR AND COMMUNITY CLASS, BY PLACE OF RESIDENCE: 1939 PLUS 1940

Age	White				Colored			
	Rural		Urban		Rural		Urban	
	Male	Female	Male	Female	Male	Female	Male	Female
0- 4	2,067	1,571	693	522	1,128	961	505	454
5- 9	165	132	49	31	74	68	35	24
10-14	142	98	50	18	102	89	42	32
15-19	311	213	77	54	202	186	97	120
20-24	428	214	110	101	239	256	154	144
25-29	361	228	145	115	296	209	175	199
30-34	375	251	188	129	242	170	229	190
35-39	425	284	252	161	282	237	251	220
40-44	466	271	289	204	350	293	288	284
45-49	552	372	450	233	434	309	353	314
50-54	698	486	537	316	565	425	431	407
55-59	914	548	668	415	564	441	390	362
60-64	1,154	824	801	503	603	493	355	383
65-69	1,499	1,020	875	713	584	422	270	279
70-74	1,530	1,136	729	669	515	344	200	204
75-79	1,364	1,131	618	706	326	244	94	115
80-84	1,140	1,058	396	618	239	178	55	57
85-89	557	592	207	290	130	115	26	22
90-94	166	226	66	120	67	68	7	14
95-99	37	52	9	24	23	28	3	2
100 and over	1	11	2	4	4	21	4	6
Unknown	6	4			10	6	4	2
Total	14,358	10,722	7,211	5,946	6,979	5,563	3,968	3,834

estimated deaths for the two years.) Eight communities in Virginia were urban by Census definition in 1940 which had not been so in 1930. For these, it was necessary to estimate the age distribution of deaths in 1939. The distribution was estimated in this way:

(1) The 1940 deaths by sex, color, and age group in each urban place were expressed

of the rate was also estimated for the middle of the two year period, *i.e.*, as of January 1, 1940. This was done mainly by linear interpolation. For each group the 1930 population was subtracted from the 1940 population. One-fortieth of this difference was taken as the increase for the three months from January 1, 1940 to the Census date, April 1, 1940. This increase was subtracted from

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the 1940 figures to get the estimated January 1, 1940 population. The procedure is summarized by the formula:

$$\begin{pmatrix}\text{Population}\\ \text{Jan. 1940}\end{pmatrix} = \begin{pmatrix}\text{Population}\\ \text{April 1940}\end{pmatrix} - \frac{1}{40}\left(\begin{pmatrix}\text{Population}\\ \text{April 1940}\end{pmatrix} - \begin{pmatrix}\text{Population}\\ \text{April 1930}\end{pmatrix}\right)$$

The eight places which changed class by Census definition were treated separately by the same method for the total population. Then the age distribution for each place was estimated for January 1, 1940 by assuming the percentage distribution to be the same as that given by the Census for April 1, 1940. These estimates were added to the urban population for January 1, 1940[1] and subtracted from the estimated rural population.

To this point ten year age groups had been used for ages of 35 and over. These were then split into the desired five year groups on the assumption that the five year groups of January 1, 1940 were in the same ratios as those of April 1, 1940. (Table 2 shows the estimated populations.)

The specific death rates by five year age groups were calculated as the final step and are presented in Table 3. These represent average annual resident deaths based on the mortality experience of 1939 and 1940 divided by estimated thousands of population on January 1, 1940. They provided a means of comparing mortality by certain classes of population, but leave as hidden classifications other important qualities such as economic and marital status.

TABLE 2. ESTIMATED VIRGINIA POPULATION FOR JANUARY 1, 1940 BY AGE, SEX, COLOR AND RURAL OR URBAN RESIDENCE*

Age	White				Colored			
	Rural		Urban		Rural		Urban	
	Male	Female	Male	Female	Male	Female	Male	Female
0– 4	66,775	64,587	24,576	23,361	23,537	23,669	8,838	9,061
5– 9	69,226	66,767	23,975	23,566	25,187	25,098	9,743	9,895
10–14	73,185	69,492	26,692	26,718	26,375	25,946	10,609	11,261
15–19	73,342	68,381	32,578	32,233	26,421	24,338	11,064	13,047
20–24	61,569	55,259	35,273	36,842	21,113	18,251	10,733	13,386
25–29	52,688	49,513	35,070	36,428	16,643	14,028	10,444	12,906
30–34	46,686	44,659	31,861	33,038	12,521	10,822	9,068	10,875
35–39	41,607	40,511	27,695	28,534	11,045	10,817	9,058	11,371
40–44	36,637	35,267	23,882	25,018	10,226	10,189	8,379	9,330
45–49	33,886	31,777	20,760	22,062	9,923	9,298	7,034	7,958
50–54	29,720	28,028	17,496	18,718	9,402	8,378	5,778	6,116
55–59	25,267	23,391	13,740	15,278	7,243	6,138	3,919	4,171
60–64	21,432	20,264	10,817	13,004	6,230	5,341	2,932	3,102
65–69	18,263	16,510	8,013	10,153	5,699	4,445	2,513	2,692
70–74	11,612	10,493	5,130	6,694	3,342	2,866	1,323	1,658
75 and over	11,127	11,221	4,429	6,679	3,320	3,330	1,011	1,468
Total	673,022	636,120	341,987	358,326	218,227	202,954	112,446	128,297

* These figures are certainly not significant beyond three digits. The extra digits are carried for convenience in tabulation.

[1] The urban population had been calculated on the basis of the 1930 areas. Cape Charles was considered as rural in 1930.

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COMPARISONS BY COLOR

Colored or nonwhite death rates are higher than white. It is obvious from Table 3 that there are apparent exceptions to this generalization. The nonwhite urban males in the 65 to 69 age group have an apparent death rate slightly below the urban white males. This may be due to overstatement of age in and that misstatements are equally divided between overstatements and understatements.[2] Then the age distribution at older ages may be seriously distorted. Thus an actual population of 2,932 nonwhite urban males 60 to 64 years of age would contribute 366 persons to the age group 65 to 69. About 314 persons in the group of nonwhites

TABLE 3. VIRGINIA DEATH RATES PER 1,000 POPULATION BY RACE, AGE, SEX, AND PLACE OF RESIDENCE*

Age	White				Colored			
	Rural		Urban		Rural		Urban	
	Male	Female	Male	Female	Male	Female	Male	Female
0– 4	15.5	12.2	14.1	11.2	24.1	20.3	28.7	25.1
5– 9	1.2	1.0	1.0	.7	1.5	1.4	1.8	1.2
10–14	1.0	.7	.9	.3	1.9	1.7	2.0	1.4
15–19	2.1	1.6	1.2	.8	3.8	3.8	4.4	4.6
20–24	3.5	1.9	1.6	1.4	5.7	7.0	7.2	5.4
25–29	3.4	2.3	2.1	1.6	8.9	7.5	8.4	7.7
30–34	4.0	2.8	3.0	2.0	9.7	7.9	12.6	8.7
35–39	5.1	3.5	4.6	2.8	12.8	11.0	13.9	9.7
40–44	6.4	3.8	6.1	4.1	17.1	14.4	17.2	15.2
45–49	8.1	5.9	10.8	5.3	21.9	16.6	25.1	19.8
50–54	11.7	8.7	15.4	8.4	30.1	25.4	37.3	33.4
55–59	18.1	11.7	24.3	13.6	38.9	36.2	49.8	43.4
60–64	26.9	20.3	37.0	19.3	48.4	46.3	60.5	61.7
65–69	41.0	30.9	54.6	35.1	51.4	47.5	53.7	51.8
70–74	66.0	54.3	71.1	50.0	77.1	60.2	76.0	61.5
75 and over	146.8	136.8	146.5	131.9	118.8	98.2	93.5	73.6

* Deaths = (1939 deaths + 1940 deaths) ÷ 2.
Populations estimated for January 1, 1940, by interpolation from Censuses of 1930 and 1940.

1940 which may have been motivated by the Social Security Act and other relief provisions. The population enumerated as of the age 65 to 69 is simply too large. This point will be discussed later.

Perhaps a more interesting exception to the generalization is the evident lower mortality of nonwhites of 75 and over. This too is a doubtful exception. There are unquestionably great errors in statements of age in these groups of very old people. An example may suffice to indicate the possible effect of these misstatements. Assume that the probability of age being stated within the correct five year age group is three-fourths for a nonwhite of 50 years or more 65 to 69 would be included in the next lower age group. This would be a net gain of 52 persons from the younger group. A similar loss of 314 persons to the age group 70 to 74 would be partly compensated by a contribution of 165 persons from an actual population of 1,323 nonwhite urban males or a net loss of 149 younger persons to the older group. This systematic error would mean that the Census populations in the high age groups are slightly younger than they

[2] Some basis for these assumptions is given by Paul M. Densen in "Family Studies in the Eastern Health District: II. The Accuracy of Statements of Age on Census Records," *The American Journal of Hygiene*, XXXII, No. 1, Sec. A. (1940), 1-38.

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are stated to be. If the same thing is true of ages stated on death certificates, then the comparison of two classes with wide differences in degree of misstatement of age is a comparison of a younger group with an older one.

A Texas study resulted in an indication of lower death rates for rural nonwhites under five than for the whites.[3] This was attributed to nonregistration of deaths in rural Texas, though it may have been greatly affected by the distribution of rural population. Virginia has been in the registration area much longer and has better death registration. Consequently, the nonwhites do not have lower deaths rates than whites for children under five.

Where nonwhites appear to have lower death rates than whites there are known biases in the data which tend to produce just that result. "It is open to question whether the lower mortality above 75 [among colored] is a genuine phenomenon due to the elimination, below that age, of a greater proportion of persons of lower vitality, or whether it is partly or wholly the result of overstatement of age in the Census by aged nonwhites."[4] Probably the elimination of persons of lower vitality is more complete among nonwhites, but the importance of misstatements of age seems much greater, so that the favorable nonwhite death rates at advanced ages would be likely to vanish if accurate age reporting were achieved. Therefore, we favor this hypothesis, "The real death rates of nonwhites are higher than those of whites at all ages." But, it should never be forgotten that nonwhite connotes differences in economic status, caste, marital status, and other attributes.

COMPARISONS BY AGE

The minima of the nonwhite death rates occur at earlier ages and are considerably higher than those of whites. Five year age specific death rates do not show this as clearly as narrower age bands would. Enumeration should be fairly good at ages of 5 to 14 and statements of age very accurate. Death registration, particularly for rural Negroes, may be poor in some cases. Death rates are very low for children of school age, which means that death frequencies are too small to permit very elaborate classification (See Table 1). The class of Virginia white urban females had only eighteen deaths for the age group 10 to 14 years in 1939 and 1940. This is the most extreme group, but it is apparent that further classification of these eighteen deaths (by size of city for example) might produce some peculiar results, because of the very small death frequencies.

The distortion of the population figures by the hope of securing old age assistance is best shown in the urban nonwhite groups of age 65 to 69. Both males and females have lower death rates at 65 to 69 than at 60 to 64. The rural nonwhites were also sufficiently influenced so that their age specific death rate charts are inflected at these ages. Theoretically, after reaching a minimum in childhood the death risk should increase with increasing age so as to produce a curve of death rates which is concave upward throughout adult life.[5] Generally speaking, the white death rates do this while the nonwhite death rates zigzag when plotted on a chart (See Charts 1 and 2) giving evidence of misstatement of age, poor allocation of deaths, etc. When deaths are not subject to irregular variation due to epidemic, war, revolutionary new medical arts, etc., they provide a basis for correcting the population figures. Estimates of the nonwhite population better than those given by the 1940 Census might be derived by such means.

We have pointed out above that survival ratios are low at advanced ages and that this makes the inaccuracies of age reporting,

[3] J. Lambert Molyneaux, "Differential Mortality in Texas," *American Sociological Review*, X, No. 1, (1945), 17-25.

[4] "United States Life Tables: 1930-1939 (Preliminary)," Washington: Bureau of the Census, July 21, 1941, p. 3.

[5] But, perhaps we should not close our minds altogether to the possibility of other minima, or points of inflection.

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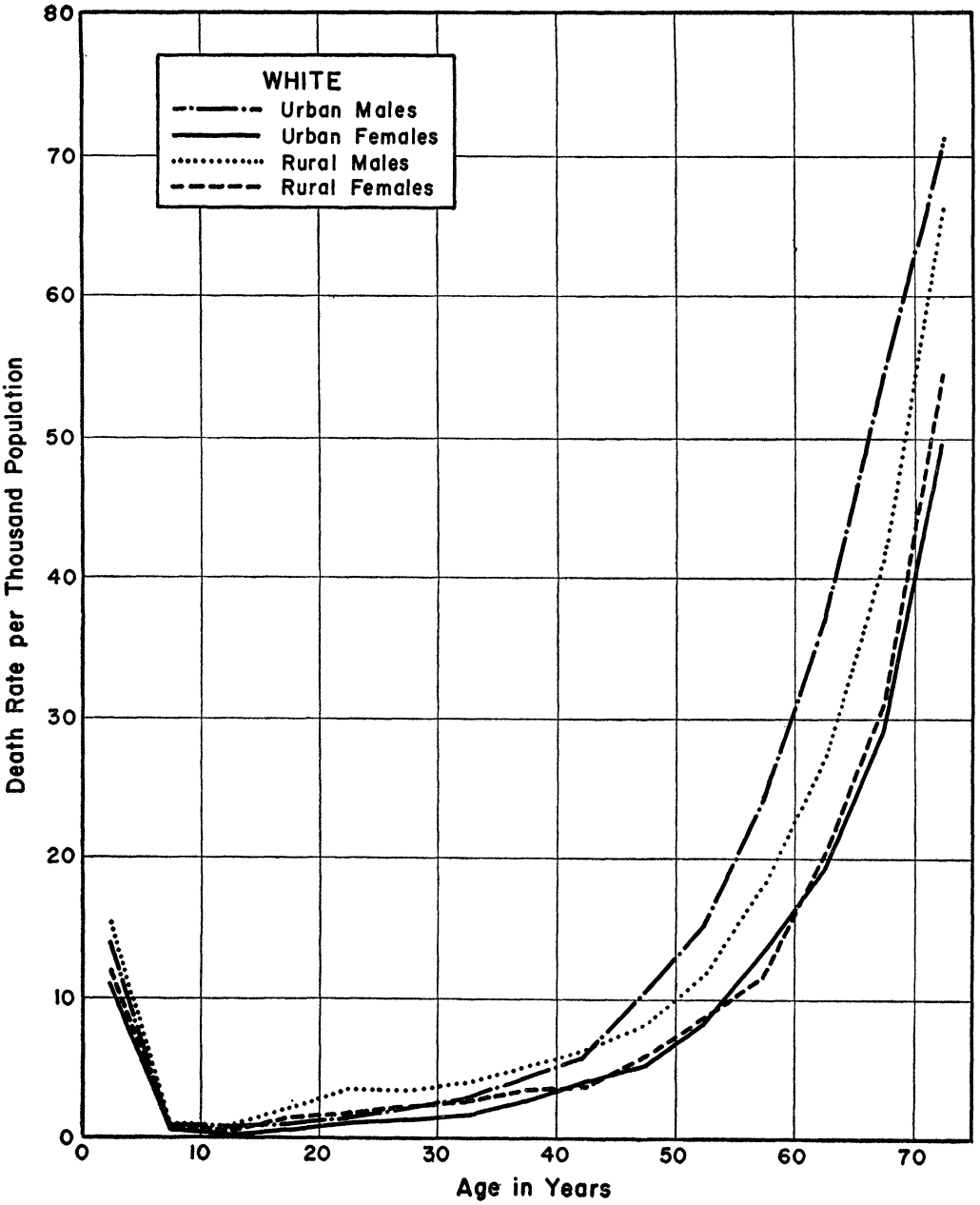


CHART 1. Virginia White Age Specific Death Rates by Sex and Rural or Urban Residence.

which are characteristic of nonwhite populations, much more important. This factor is probably more important than any other in holding down the apparent death rates of nonwhites of advanced age.

The discussion of age differences in infancy and early childhood is left to the section on residence differences because residence groups, rather than age groups, provide the most significant comparisons.

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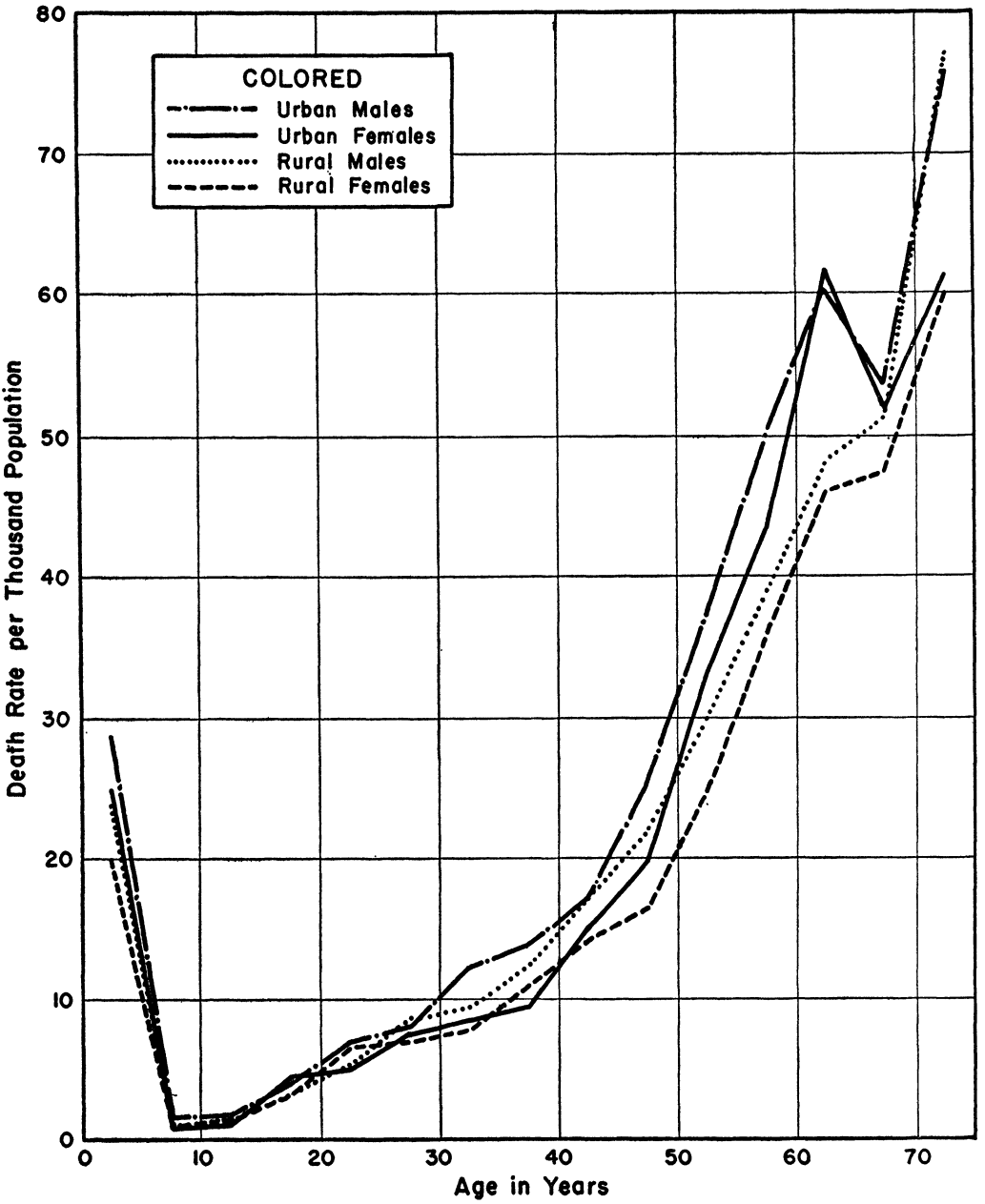


CHART 2. Virginia Colored Age Specific Death Rates by Sex and Rural or Urban Residence.

COMPARISONS BY SEX

Females have lower death rates than males. There are no exceptions to this rule for whites as far as the specific death rates by five year age groups in Virginia are concerned. In the case of the nonwhites of 15 to 19 years and of 20 to 24 years in rural places the females have death rates equal to or greater than the males. This is due to high fertility combined with adverse con-

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ditions of sanitation, medical care, etc. The analysis of these differences will require further detailed study, particularly of deaths due to causes connected with tuberculosis and childbearing. A more refined analysis would probably show that there are ages at which certain classes of white females have higher death rates than males. This is most likely in the case of married urban whites of 15 to 19 years among whom there are a great many women having first births before reaching the age of minimum maternal mortality.[6]

Illegitimate births are also adverse to the survival of the mother. The nonwhites have much higher numbers of illegitimate live births per 1,000 live births than the whites. For girls of 15 to 19 years in Virginia the 1943 nonwhite rate was 415.4 illegitimate live births per 1,000 live births, while for the whites it was only 66.5 illegitimate births per 1,000 live births.[7] The illegitimacy rate for girls 15 to 19 was, therefore, six and one-fourth times as high for colored as it was for whites.

The qualified conclusion is that when females appear to have higher death rates than males it is due to childbearing under conditions which produce high maternal mortality. Wider use of the sulfonamides may eliminate enough of the risk of maternity to reduce all female rates below those of males in the near future.[8]

Another group of nonwhite females in Virginia which has a higher death rate than the corresponding males is the urban group of 60 to 64 years of age. Reference to Chart 2 will show that the overstatement of age by nonwhite urban females who belong to this group is more extreme than that for males. This would have the effect of reducing the population of 60 to 64 and thus increasing the death rate, while increasing the population of 65 to 69 and thus decreasing the death rate for that group. The effect on nonwhite urban females is so severe in Virginia that those of 70 to 74 years appear to have death rates slightly lower than those of 60 to 64 years!

The race differential in misstatement of age is much more extreme than the sex differential. White death rates give little evidence of distortion at ages of 65 to 69. The distortion among married nonwhite is not very great.

For the United States, the divorced urban nonwhite males have death rates of 85.8 at 60 to 64 years and 75.7 at 65 to 69. For widowed urban nonwhite females the death rate of 51.1 at 60 to 64 drops to 45.9 at 65 to 69 years. In rural places overstatement of age from security motives is most evident for single and widowed nonwhite females.[9] In the age group of 60 to 64 years 55.5 per cent of Virginia's nonwhite urban females are classified in the 1940 Census as widowed while only 20.9 per cent of the males in the group are so classified. In the age group 65 to 69 these percentages are 66.2 and 26.9. Since the widowed are the class exhibiting the most extreme overstatements of age among urban nonwhite females in the United States and there are so many more widowed females than widowed males in Virginia, it must be that the heavy concentration of nonwhite widows in Virginia cities is responsible for the strange behavior of the nonwhite urban death rates.

COMPARISONS BY RESIDENCE

Since the time of John Graunt (1620-1674) the generalization that the urban is higher than the rural death rate has been almost universally accepted. We have reached

[6] See "Mortality by Marital Status by Age, Race, and Sex, Urban and Rural, United States, 1940," *Vital Statistics—Special Reports,* Vol. XXIII, No. 2, July 3, 1945.

[7] The figures quoted are taken from "Illegitimate Live Births by Race, United States, 1943," *Vital Statistics—Special Reports,* Vol. XXI, No. 15, November 15, 1945.

[8] Puerperal death rates per 10,000 live births for the United States dropped from 37.6 in 1940 to 24.5 in 1943. More than 57¼ per cent of this life saving was from death caused by puerperal septicemia. See "Death From Puerperal Causes by Race and by State, United States, 1943," *Vital Statistics—Special Reports,* Vol. XXI, No. 5, May 8, 1945, p. 88.

[9] See "Mortality by Marital Status by Age, Race, and Sex, Urban and Rural, United States, 1940," *Vital Statistics—Special Reports,* Vol. XXIII, No. 2, July 3, 1945.

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a time when it becomes necessary to qualify this proposition. In the case of Virginia white females, it appears that urban death rates are lower than rural until the menopause. The five year age groups 40 to 44, 55 to 59, and 65 to 69, show higher death rates for urban than for rural white females but all other ages show an advantage for the urban places. The urban white males also appear to have better mortality experience than the rural up to about age 45, though not after that age. The ages at which rural whites do seem to have an advantage are those at which age reporting becomes less reliable. It may be that overstatement of age is more common among the rural whites of advanced years and, hence, gives them lower overt death rates, but it would be hard to prove this.

In the case of the nonwhites there are also interesting exceptions to the old rule. Females in the age groups 5 to 9, 10 to 14, 20 to 24, 35 to 39, and 75 and over, have lower death rates in urban places. Nonwhite males 25 to 29, and 70 or over, have lower death rates in urban places. There are also significant differences in economic and marital status which might reduce the differences between urban and rural rates in other age groups.

Table 3 shows that the colored under five years of age have higher death rates in urban than in rural places. Deaths under one year are by far the most important in this age group. Table 4 shows that the death rates for the colored infants are much higher in all sizes of cities. The result is that, no matter how the nonwhite population is distributed in the cities, the rural population has a decided advantage. For the white population under five, the smaller the proportion living in places of 10,000 population or less, the lower the death rates would be. It happens that almost twice as many whites proportionately (24.42%) as colored (12.25%) live in places of 2,500 to 10,000

TABLE 4. VIRGINIA SPECIFIC DEATH RATES BY AGE, RACE, AND POPULATION-SIZE GROUPS, BY PLACE OF RESIDENCE: 1940*
(Exclusive of stillbirths. Rates are the number of deaths in a specified group per 1,000 population of that group, enumerated as of April 1, 1940)

Age	White				All Other			
	Rural	Cities of 2,500–10,000	Cities of 10,000–100,000	Cities of 100,000 or more	Rural	Cities of 2,500–10,000	Cities of 10,000–100,000	Cities of 100,000 or More
All ages	9.2	11.5	9.9	10.4	14.8	18.8	17.3	16.0
Under 1 year	64.1	70.5	52.9	53.4	100.8	137.9	125.3	113.3
1– 4	2.9	3.3	2.6	2.8	4.7	4.5	4.2	5.5
5–14	0.9	1.1	0.6	0.9	1.7	1.6	1.7	1.4
15–24	2.0	2.0	1.6	1.6	4.7	5.9	5.5	6.4
25–34	2.8	3.1	2.3	2.1	7.7	9.2	9.3	9.0
35–44	4.3	4.8	4.4	4.5	12.7	17.3	13.7	12.3
45–54	8.1	12.2	10.9	9.9	22.7	28.2	29.9	31.4
55–64	18.6	24.9	23.7	22.6	42.7	59.2	54.2	50.0
65–74	44.1	61.7	55.4	53.9	58.3	66.2	65.7	62.7
75–84	118.3	150.7	126.8	130.5	99.7	97.1	109.6	57.9
85 and over	261.1	287.1	250.8	290.6	181.4	137.0	109.7	52.6

* From Table VIII.—*Vital Statistics of the United States*, Supplement 1939–1940, Part III, p. 78.

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population. This distribution, which is favorable to lower urban rates for colored is not sufficiently favorable to overcome their higher death rates. The whites, on the contrary, are sufficiently concentrated in urban places above 10,000 population to give them urban death rates under five years which are slightly better than the rural rates.

Nonregistration of death may be of considerable importance in some areas, particularly for nonwhite rural farm infants under one. If so, then the nonwhite rural rates for this group would be affected because 64.4 per cent of the colored rural population under five is farm population as compared to only 52.3 per cent of the white.

The urban colored under five in Virginia have death rates more than twice as high as the urban whites. Since better than one quarter of the State's population is affected by this fact, it is obvious that expectation of life in Virginia could be most quickly increased by a reduction of this difference. The greatest difference under one year exists where the white rate is lowest, in places of 10,000 to 100,000 population. But, some action should also be devised to reduce the rural color differences which involve almost three fourths of the Virginia population.

If sex composition is neglected, rural and urban rates can be compared by means of Table 4 which provides breakdowns by size of city. This table indicates that both whites and nonwhites under one year have their worst mortality experience in cities of 2,500 to 10,000. In Virginia both whites and nonwhites appear to have their best mortality experience in the age group of one to four years in places of 10,000 to 100,000, but the number of deaths is small. Whites in Virginia have their highest death rates in small cities of 2,500 to 10,000 up to 85 years, at which age these places seem to have a slight advantage only over the largest cities. There seem to be no other clear cut patterns. The relationships between these death rates would vary widely in other states, but for the whole United States the higher force of mortality in the smallest urban places holds pretty well through life for nonwhites and up to age 45 for whites.

The comparison of age specific death rates has an important danger in the possibility of over emphasizing differences in rates where only a few deaths are involved. For this reason it is well to glance at the age-adjusted[10] rates given in Table 5. The crude rates range from the best of 8.30 for white urban females to the worst of 17.64 for nonwhite urban males. Age-adjustment inter-

TABLE 5. VIRGINIA CRUDE AND AGE-ADJUSTED DEATH RATES, 1939–1940

	Crude Virginia	Age-Adjusted	
		Virginia Population Standard	United States Population Standard
White urban males	10.54	11.04	12.52
White urban females	8.30	7.64	8.67
White rural males	10.67	10.44	11.67
White rural females	8.43	8.13	9.15
Colored urban males	17.64	18.64	20.43
Colored urban females	14.94	16.04	17.58
Colored rural males	15.99	16.80	18.52
Colored rural females	13.71	14.64	16.08

changes the ranks of urban and rural white males, indicating that white urban males have an age distribution very favorable to low crude death rates. Other ranks are not changed by the operation of adjusting for age. The indicated rank order from best mortality experience to worst is white urban females, white rural females, white rural males, white urban males, colored rural females, colored urban females, colored rural males, and colored urban males. The advantage of urban residence for white females is clear, but about 79 per cent of them were living in cities of over 10,000 population in 1940, a residence distribution favorable to

[10] Age-adjusted or standardized rates are theoretical rates computed on certain assumptions. Assuming that the known age specific death rates for a given class are operating on a population with a certain age distribution as standard (Virginia and United States in the present case) there is a given number of deaths which would result. This number divided by the standard population (all ages) gives the age-adjusted death rate.

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low death rates. The age distribution of all these classes in Virginia is more favorable to low crude death rates than is that of the United States as a whole. The colored in Virginia also have a favorable age distribution relative to the whole state, and, hence, to the white. Except for the urban males, the whites have age distributions unfavorable to low crude death rates relative to the total population of Virginia.

Conclusions

For the most part the age specific death rates of Virginia confirm generally accepted ideas on the incidence of deaths. The propositions which follow are therefore mainly restatements of the accepted generalizations. However, no one could make a study of age specific death rates in any individual state without having departures from accepted rules forced upon his attention. That is the source of interest in the work and the reason why the development of thought on the subject is partly dependent on the study of vital phenomena in small geographic areas. The words in which these propositions are stated are not, of course, a substitute for study of the tables. The essence of the matter is the magnitudes of the differences, which can be seen only by comparing the numbers in the death rate tables.

1. Colored death rates are higher than white. Where this appears not to be true it is known that bias exists in the data. It is also known, however, that old persons who have survived bad environments do tend to have better vitality, i.e., that sanitary and medical protection from infection and contagion tend to keep alive more persons susceptible to degenerative disease, so that certain classes of colored have lower death rates from certain causes than the corresponding classes of whites.
2. Colored minimum death rates occur at lower ages than those of whites. Combining these two propositions would give the conclusion that white minima are lower than those for colored, which is the empirical fact. Minimum death rates occur at ages where there are so few deaths that there can be considerable random error, however.
3. Females have lower death rates than males. Possibly, there would be no exceptions to this rule if allowances were made for deaths from puerperal causes. But, with these deaths included, the young colored women have higher death rates than males.
4. The highest death rates by community class are found in urban places of small size. When sex, as well as race, is considered, it is found that white urban females have a decided advantage in mortality experience. These may be transition phenomena due mainly to the differences in the speed with which sanitary devices are adopted. Low densities of population in small towns and rural areas rule out certain expensive types of water supply, sewage disposal, and medical services. Large urban centers can bear the institutional load of complicated public health services. Sanitary and medical facilities for the colored population usually lag decades behind those for whites. If communities were classified by the presence or absence of approved facilities, it would probably be found that these are the essential attributes rather than size of place. Sanitary practices as now developed are mainly those applicable to large aggregations of population in urban centers.

The practical suggestions for Virginia which may be derived from this research are that special efforts should be made to improve health services in small urban places, that better maternal care should be available in rural places, and that instruction in infant care is badly needed by colored people. Particular efforts should be made to bring colored death rates down nearer to those of whites, for the benefit of the whites as well as the colored, since microbes do not respect caste lines.

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